

THE MARKET LAYER

The Mandate Market

How capital, assets and credit find each other on attested data: the matching engine beneath the autonomous market, mechanically.

THE STARTING POINT

An order book needs one thing real estate never has

WHAT A CLOB ASSUMES

- One identical, fungible instrument per book
- Both sides comparable on a single axis: price
- Price-time priority resolves every tie
- Matching is deterministic; nobody exercises judgment
- Speed is the competitive variable

WHAT REAL ESTATE HAS

- Every asset is a unique instance; no two are substitutes
- Value is a vector: yield, term, credit, location, capex
- Buyers rank assets and sellers rank buyers, differently
- Certainty of close matters as much as price
- Old data is worse than slow data

The right frame is a **two-sided ranked matching market**: standing intents on both sides, each scoring the other, cleared by a matching rule rather than by price-time priority. Debt joins it as a sub-market that runs per pair.

THE OBJECTS

Three object types, not three sides

01 · ASSET

The instance object

The only thing that exists once. An attested vector of dimensions with lineage, plus a seller instruction: ask, reserve, timing, conditions.

Produced by Layer 01 and 02. Changes when its state changes, not when the market does.

02 · BUY MANDATE

A standing parametric intent

Hard filters, a utility curve per dimension, weights, floors, a tie-break order and a return target. It scores assets. It does not name one.

Rests in the book until filled, amended or withdrawn. Submitted by a human or a permissioned agent.

03 · CREDIT BOX

A conditional standing intent

A lender's box: asset types, LTV cap, DSCR floor, spread grid, sponsor tier, capacity. It scores the asset conditional on who is buying it.

Size shrinks as it fills. The closest thing in the system to a resting limit order.

Two books score the asset. Mandates score it directly; credit boxes score it given the mandate. So the match stays bipartite, asset against mandate, with debt attached to the pair.

The seller is a fourth scorer. Each asset carries its owner's ranking of bids: price, certainty, speed, conditionality. That ranking is what makes the match stable.

OBJECT 01 · THE ASSET

The asset enters the book as an attested vector

Dimension	Shape	Attested from	Refresh
Ask price	ceiling	Seller instruction	on change
NOI	monotone, more	Rent roll and ledger · Proof of State	30 days
Cap rate	monotone, more	Derived: NOI over price	with NOI
WALT	monotone, more	Lease register	lease event
Tenant credit	tiered	Rating feeds, arrears monitoring	quarterly
Vacancy	monotone, less	Rent roll	30 days
Location and class	categorical	Structuring engine tiering	annual
Capex backlog	monotone, less	Technical DD, monitoring	on inspection
In-place debt	structural	Loan register, lender consent status	on change

SHAPE DECIDES THE MATH

Most dimensions are monotone: more yield, more term, more credit is simply better. Only a few are target-shaped (lot size) or categorical (class). Distance-to-target is the wrong metric for almost all of them.

EVERY VALUE CARRIES LINEAGE

Each field has a source, an attestation timestamp and a confidence. The scoring engine reads the confidence; the credit box refuses to fire without it.

Illustrative schema. Refresh windows are policy choices per dimension.

OBJECT 02 · THE MANDATE

A mandate is a scoring function with five parts

01

Hard filters

Binary. Fail one and the asset never enters this mandate's book. Class, geography, lot size, structure.

02

Utility curves

One per soft dimension, mapping the raw value to 0 to 1. Linear, capped, S-shaped or stepped.

03

Weights and floors

Weights combine the utilities. Floors stop one strong dimension masking a fatal one.

04

Tie-break order

Lexicographic fallback when scores are within tolerance. Replaces time priority.

05

Return target

Levered return the buyer requires. With the debt quote, this is what turns a score into a bid price.

MANDATE M-0417

illustrative

Hard: logistics · Germany · lot €10 to 25M · cap rate at least 5.25% · no in-place debt requiring consent

Cap rate: linear 5.25% to 7.0%, capped · **WALT:** linear 3y to 8y · **Tenant credit:** tiered IG 1.0, BB 0.6, B 0.2 · **Vacancy:** linear 0 to 10%, less is better

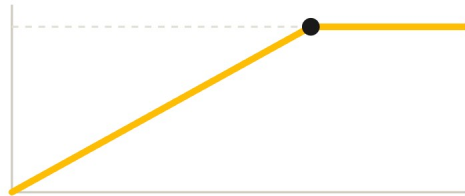
Weights: cap 45 · WALT 25 · credit 20 · vacancy 10 **Floors:** vacancy at most 5% · credit at least BB

Tie-break: longer WALT, then lower capex **Target:** 12% levered IRR, 5 year hold

"Modes" are presets of parts 02 to 04. Yield-hungry, capital-preservation and waterfall are one engine with different parameters, not three engines.

OBJECT 02 · UTILITY, NOT DISTANCE

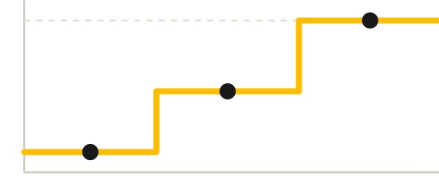
Each dimension maps to a utility between 0 and 1

MONOTONE, CAPPED

Cap rate, WALT, credit. More is better up to a point past which the buyer stops paying for it. Below the floor the utility is 0 and the asset drops.

TARGET-SHAPED

Lot size, location radius. The only dimensions where distance-to-target is right. A plateau, not a point: €12M and €18M are equally fine for a €10 to 25M mandate.

STEPPED, CATEGORICAL

Asset class, building grade, tenant tier. The structuring engine turns "Class A" and "needs light refurbishment" into tiers first; the mandate assigns a value to each tier.

Why not Euclidean distance. Raw units are incommensurable (euros against percentage points), monotone dimensions have no centre to be far from, and a single distance hides which dimension failed. Utilities are normalised, directional and explainable.

Confidence multiplies in. Each raw value arrives with an attestation age. Utility is scaled by a decay factor, so a 6.2% cap rate attested last week outranks 6.4% attested five months ago.

Curves are illustrative shapes, not calibrated values.

From asset and mandate to a ranked position, in five steps

01**Filter**

Apply hard constraints. A miss rejects the pair outright and is logged with the failing field, so the rejection is auditable.

02**Score**

Run each soft dimension through its utility curve, then scale by the field's confidence and attestation age.

03**Combine**

Weighted sum, after every floor is checked. Any dimension under its floor sets the score to zero regardless of the rest.

04**Pareto check**

Flag assets dominated on every dimension by another in the same book. Shown, not hidden: the buyer sees the frontier beside the score.

05**Rank**

Place in the mandate's book by score, tie-break order on near-ties. Price is set later, from the debt quote and the return target.

THE SCORE

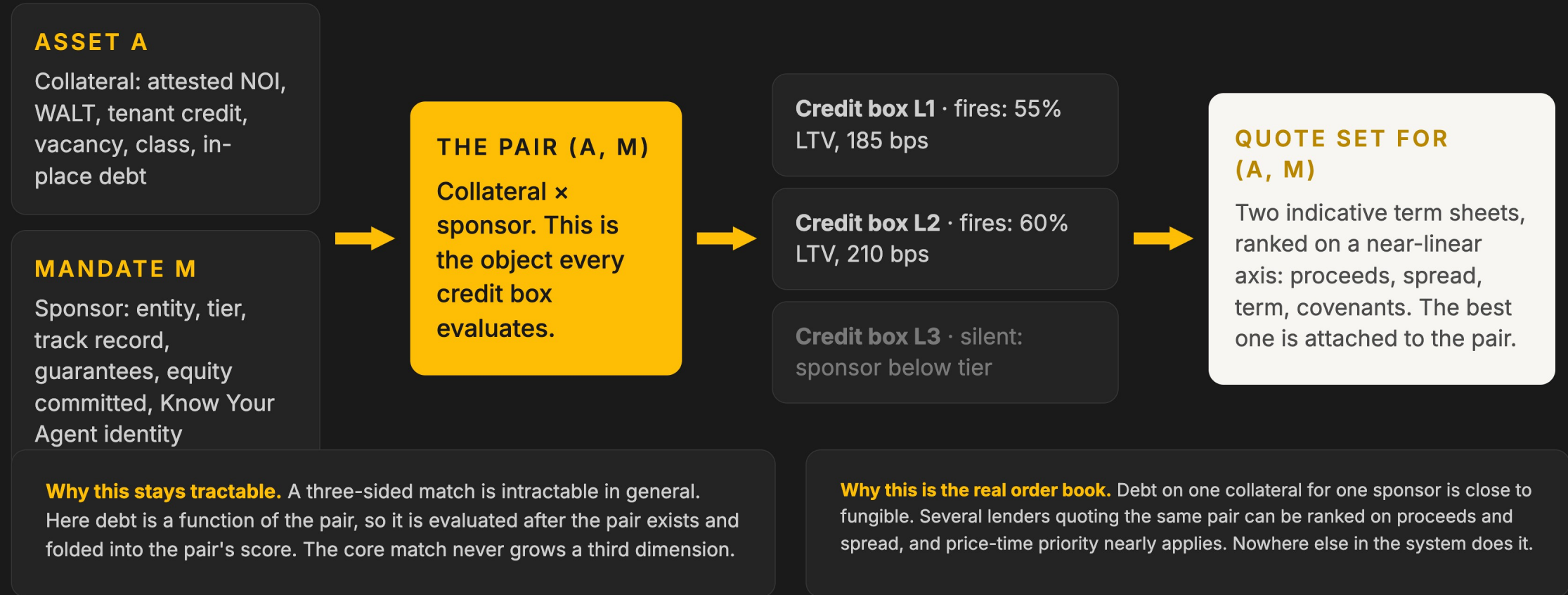
$$S(a, m) = \sum w_i \cdot c_i(t) \cdot u_i(a_i) \quad \text{if every } u_i \geq \text{floor}_i, \text{ else } 0$$

w : mandate weight · u : utility curve · $c(t)$: confidence decay from attestation age · floors from the mandate.

WHY FLOORS AND A FRONTIER

A weighted sum alone lets a 7% yield hide a 20% vacancy, and adding a new asset can reorder existing ones. Floors stop the first; showing the Pareto frontier next to the score makes the second visible instead of silent.

Lenders quote neither the asset nor the buyer. They quote the pair.



Quotes shown are illustrative.

A credit box is a resting order whose size shrinks as it fills

Field	Illustrative value	Role in the engine
Asset types, geographies	Logistics, light industrial · DE, NL	Hard filter on the asset
Sponsor tier	Tier 1 and 2 only	Hard filter on the mandate
LTV cap	55%	Proceeds constraint, price-linked
DSCR floor	1.30×	Proceeds constraint, NOI-linked
Debt yield floor	7.5%	Proceeds constraint, NOI-linked
Spread grid	150 bps to 50% LTV, 185 to 55%	Price of the quote by band
Term, amortisation	5 years, 1% per year	Debt constant for the buyer's model
Capacity, concentration	€120M left · max 15% one tenant	Remaining size; decrements on fill
Attestation requirement	NOI attested within 60 days	Gate: no attestation, no quote

INDICATIVE FIRST, FIRM ON REQUEST

The box fires automatically with indicative terms. A firm term sheet is a triggered request for quote on the pair, answered by the lender's agent under its own permissions and logged as Proof of Agency.

THE ATTESTATION GATE

A buyer tolerates a fuzzy vector and diligences later. A credit box cannot: it prices off NOI, and unattested NOI is not a number it can underwrite. The debt book is where Layer 02 is binding, not optional.

Values are illustrative. Field set follows standard senior lending parameters.

Price depends on debt and debt on price. The grid breaks the loop.

- 01** **Lenders quote constraints, not a price**
 Proceeds at any price P : $D(P) = \min(LTV \cdot P, NOI \div (DSCR \cdot k), NOI \div DY)$
 k is the debt constant from the grid: rate plus amortisation.
- 02** **Equity is what remains**
 $E(P) = P - D(P)$. Levered return $r(P)$ falls as P rises, because both the equity cheque grows and the yield on cost falls.
- 03** **Solve for the maximum price**
 P^* = the largest P with $r(P) \geq \text{target}$. $D(P)$ is piecewise linear and $r(P)$ is monotone, so P^* is unique and a bisection finds it in a few steps per lender quote.
- 04** **The bid is the best P^* across quotes**
 $\text{Bid} = \min(P^*, \text{ask})$, financing attached. A more generous credit box raises P^* directly: debt competition shows up as equity price.

WORKED EXAMPLE

illustrative

Attested NOI	€800k
Credit box L1: LTV cap · DSCR floor · debt yield floor	55% · 1.30× · 7.5%
Debt constant k (rate plus amortisation)	6.5%
Candidate price P (cap rate 6.15%)	€13.0M
Proceeds by LTV · by DSCR · by debt yield	€7.15M · €9.47M · €10.67M
Binding constraint: LTV → $D(P)$	€7.15M
Equity $E(P)$ · debt service · cash to equity	€5.85M · €465k · €335k
Year-one cash yield on equity	5.7%
Against a 12% levered IRR target, P^* solves from here	$\text{bid} = \min(P^*, \text{ask})$

At €13.0M the LTV cap binds. Above roughly €17.2M the DSCR test would bind instead; the kink is where the grid's band changes and the spread steps up.

Numbers are a worked illustration, not a valuation.

The seller scores bids on more than price, and can prove it

Dimension	What the engine reads	Where it comes from
Price	Bid P* net of conditions and costs	The bid
Certainty of close	Firm term sheet attached, or indicative, or none · equity evidenced	Credit box status · KYA identity
Speed	Time to exchange: diligence already run on attested data, or from scratch	Mandate's diligence permissions
Conditionality	Subject-to clauses remaining: financing, planning, board, technical	Bid terms
Counterparty record	Closed-to-bid ratio, retrades, withdrawals	Platform history, attested
Structure fit	Assumes in-place debt, accepts vendor terms, share or asset deal	Bid terms · loan register

FINANCING IS THE CERTAINTY SCORE

The financing contingency is the single largest source of failed closes. A firm term sheet from a credit box, attested and permissioned, converts directly into a higher seller score. A higher headline bid with no debt attached can rank below it.

SAME MACHINERY, OTHER DIRECTION

The seller's ranking is a mandate too: filters, utilities, weights, floors. A fund with a redemption deadline weights speed; a family office weights price. It rests with the asset and is never shown to bidders.

Clearing is deferred acceptance, not price-time priority

01 Mandates propose

Each mandate proposes to the top asset in its book, at its computed bid P^* with the best debt quote attached.

02 Assets hold, tentatively

Each asset keeps the proposal its seller's ranking prefers and releases the others. Nothing is final yet.

03 Released mandates move down their book

A released mandate proposes to its next-ranked asset. Its debt quote is re-evaluated, since the pair changed.

04 Stop when nobody moves

The held pairs are a stable set: no asset and mandate that are not matched would both prefer each other over what they hold.

WHY STABILITY IS THE GOAL

A best-effort engine sends one hot asset to twenty mandates and leaves nineteen to retrade. A stable set has no blocking pair, so no participant can improve by going around the platform. That is the property a market needs before agents run it unsupervised.

CONTINUOUS, NOT ONE-SHOT

Real estate has no closing bell. The rule runs as a standing process: every book change (new asset, amended mandate, credit box fill, attestation update) re-runs the affected pairs. A held pair converts to a transaction only when both sides confirm, inside a hold window.

Proposer advantage. The proposing side gets its best stable outcome. Mandates propose by default; a seller-led process for a marketed asset flips it.

The buy side is not a buyer. It is an assembled capital stack.

STACK FOR ASSET A

share · quoting intent

Senior	55%	Credit box L1, bank
Mezzanine	15%	Credit box L4, debt fund
Preferred	10%	Pref equity mandate
Common	20%	Buy mandate M-0417

Each tranche is a quote from a different standing intent, conditional on what sits above and below it. The mezz quote depends on senior proceeds; the common return depends on both. Assembly is the same root-finding as slide 10, run tranche by tranche from the top.

A MATCH IS ASSET AGAINST STACK

The equity mandate remains the proposer and the anchor, but what it proposes is a fully assembled stack with every tranche's terms attached. The seller scores the stack: total proceeds, certainty of each layer, and time to assemble the rest.

IN-PLACE DEBT IS A PRE-FILLED TRANCHE

Assumable debt enters the stack as a senior tranche with fixed terms and a consent flag. Mandates that will assume it skip the senior quote entirely and inherit its constraints; mandates that will not pay the break cost in their model.

PROGRAMMABLE SECURITISATION IS THE REVERSE

Once stacks are assembled from attested quotes with lineage, slicing a senior tranche into a pool, or re-bundling mezz across assets, is the assembly operation run backwards on the same records. Not a separate product.

Nothing in the book is stable. Four events re-run the engine.

ASSET STATE

A new attestation: rent roll refresh, lease signed or broken, arrears, inspection. Re-scores this asset in every mandate book it sits in; re-quotes every held pair.

MANDATE CHANGE

A weight, floor or target moves. Re-scores that mandate's whole book. Thousands of assets, milliseconds. The engine is not the bottleneck; the data is.

CREDIT BOX FILL

A close decrements capacity or hits a concentration limit. Every pair that leaned on that box loses its quote and drops to the next lender. Bids move; held pairs may release.

RATES AND GRIDS

A lender reprices its spread grid or the base rate moves. Every P^* in the system shifts at once. This is the closest the book comes to a market-wide tick.

DECAY REPLACES TIME PRIORITY

In an order book the earliest order wins. Here the freshest attestation wins. Each field's confidence $c(t)$ falls from 1.0 at attestation toward a floor over its refresh window, then to 0 past a hard limit. A seller who stops attesting watches the asset sink in every book, without anyone acting.

HOLD WINDOWS AND HYSTERESIS

A held pair is protected from re-ranking for a hold window so that a minor attestation update cannot break a match in progress. Outside the window, re-ranking needs the score to move past a tolerance band before a pair releases. Both are mandate-level settings, both are logged.

Three ways to run the books. Institutions will want the third.

A · OPEN

Both books visible

Mandates browse assets; sellers see resting mandates. Maximum discovery, maximum leakage. Every off-market asset becomes marketed the moment it enters; every mandate reveals its strategy.

Fits: retail fractions, small lots, auction-style disposals.

B · ONE-WAY

Assets visible, mandates dark

The conventional broker model, automated. Sellers list; mandates score privately and surface only as bids. Buyers keep their strategy; sellers still broadcast.

Fits: marketed processes with a defined bid deadline.

C · DOUBLE-BLIND

Platform holds both books

Neither side sees the other. The engine reveals a pair to both parties only when mutual scores clear a threshold, inside a hold window, with debt indicated. Off-market stays off-market until there is a reason not to.

Fits: institutional, off-market, portfolio and secondaries.

What double-blind requires. The platform must be trusted to hold both books without trading on them. Proof of Permission and Proof of Agency are what make that claim checkable: every read of a mandate or an asset by any agent, including the platform's own, is on the record.

Disclosure is per object, not per platform. A seller can list one asset open and another double-blind; a mandate can be dark for logistics and open for a disposal. The engine treats visibility as one more attribute with permissions.

IN THE STACK

The market is a Layer 03 solution that only the three layers make possible

01 · INGESTION AND STRUCTURING

Produces the asset vector: one schema, tiered categoricals, derived fields. Without it there is nothing for a utility curve to read.

02 · VALIDATION ENGINE

Supplies $c(t)$ for every field, gates the credit boxes, records every read under Proof of Permission and every bid under Proof of Agency. Makes double-blind auditable.

03 · THE MATCHING ENGINE

Books, scoring, pair quoting, deferred acceptance, hold windows. Mandates and credit boxes are submitted by identified agents under Know Your Agent. Delivered inside live operations first.

01 List

Asset enters as a vector with a seller mandate; visibility set per object.

02 Diligence

Scoring on attested fields; agents drill into lineage only on held pairs.

03 Match

Deferred acceptance across both books; stable pairs revealed in a hold window.

04 Finance

Credit boxes already quoted the pair; firm terms on request, stack assembled.

05 Settle

Both confirmations close the pair; title, token and every tranche transfer atomically.

THE CLAIM

Matching is a data outcome

The engine is not the hard part. Utilities, pair quotes and deferred acceptance are known mathematics. What no one has had is an asset that arrives as an attested vector and a credit box that can trust it.

Discussion

01 Disclosure model to launch with

Double-blind for institutional first, or one-way on marketed assets to seed volume.

02 Who proposes

Mandates by default, sellers on marketed processes. The proposing side gets the better stable outcome.

03 The credit box standard

The schema lenders submit against. Whoever defines it defines the debt book. Start with the lenders already on the platform.

04 First book to run live

The 11,000 units on platform and mandate are a closed market with attested data on both sides. Run the engine there before any external counterparty.